

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks, surfacing **86,387** duplicate records and a **47%** null rate to establish the source-of-truth for a downstream monitoring data platform.
- Designed a distributed time-series anomaly detector, benchmarking algorithms to select an ML model handling multi-layer seasonality across **10+** workflows and **30+** countries.
- Shipped SQL and MLOps pipelines automating weekly retraining and hourly model inference.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend serving **100k+** datapoints from fitness devices, designing database schemas to unify formats across **20+** distinct data sources.
- Implemented asynchronous batch processing and storage workflows on GCP, yielding a **35%** execution speedup and a **60%** reduction in redundant database writes.
- Automated environment setup and data seeding via Bash scripts, cutting testing time up to **80%**.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python/NumPy pipelines for a **15%** throughput boost on **1M+** weekly events.
- Implemented advanced data structures for visualization and analytics modules, driving **20%** faster reporting and improved data integrity across shipment dashboards.
- Enhanced workflow efficiency **15%** through Panels-based UI/UX refinements and user testing.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline analyzing LLM embedding manifolds (Llama-2, DeepSeek-V2), building weight-matrix extraction utilities to shrink **500GB** checkpoints to **13GB** localized manifolds.
- Optimized placement scripts and preprocessing for a **30x runtime reduction** (30 min to <1 min), enabling large-scale manifold analysis on resource-constrained **8GB RAM** systems.
- Applied **Rent's Rule** and Louvain clustering to identify model backbones, raising Inter-Cluster Network (ICN) density from **0.18%** to **13.25%** in Llama-2-70B.

PROJECTS

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* 

Aug. 2025 – Dec. 2025

- Engineered a full-stack platform on **AWS** with **20+ Django REST endpoints** for coach-student scheduling, real-time tracking, and payments, secured with authentication and session logic.
- Tuned **SQLite** through strategic indexing and query optimization for multi-user scalability, backed by a **CI/CD pipeline** with automated testing for **99.9%** deployment reliability.

MovieRecapinator | *Java, Spring Boot, Maven, JavaScript, IMDb API* 

April 2025

TECHNICAL SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL (Postgres/SQLite), Bash, HTML/CSS

Development: Django, Spring Boot, Node.js, FastAPI, Flask, REST APIs, WebSockets, CI/CD, JUnit, PyTest

AI & Cloud: AWS, GCP, Docker, Git, GitHub Actions, PyTorch, TensorFlow, Gemini/OpenAI API, Claude Code, Linux